

EMDLAB finite element package

Electrical Machines Design Laboratory

1 Introduction

1.1 Overview

EMDLAB is an open-source numerical package developed in the MATLAB environment for the design and analysis of electrical machines, including motors, generators, transformers, and actuators. The package provides a transparent and reproducible tool for academic research and education. Its open-source nature encourages collaboration and facilitates easy extension of the code, making it particularly valuable for method development and academic use.

The package is organized as a library of MATLAB objects. Depending on your specific design needs, you can select the appropriate modules to obtain your desired results. With EMDLAB, you can also develop customized, standalone software tailored to your applications.

1.2 Setup Instructions (Windows 64-bit)

1. Download the emdlab-win64.zip file.
2. Extract the zip file and place the "emdlab-win64" folder in the "C:" directory, without changing the folder name.
3. To use the EMDLAB package in your MATLAB code, add the following line at the beginning of your mfile:

```
--->> addpath(genpath('C:\emdlab-win64'));
```

1.3 How to install EMDLAB from GitHub? (Follow the video link)

<https://github.com/EMDLAB-Package/emdlab-win64>

https://youtu.be/ifwybm4r2_0

1.4 For academic use, cite this paper

<https://www.sciencedirect.com/science/article/pii/S2352711025004121>

1.5 How to use Gmsh as a mesh generator in EMDLAB

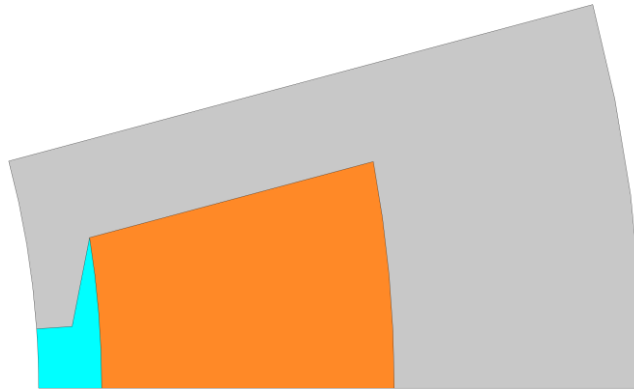
In addition to the built-in EMDLAB mesh generators, it is also possible to use Gmsh as an external mesh generator.

1. Install Python.
2. Install Gmsh -> you can use: pip install gmsh
3. Set python.exe path in the pyPath.txt file located in "C:\emdlab-win64\geometry\pyPath.txt"
4. When using the .generateMesh method, use 'gmsh' as input: .generateMesh('gmsh')

2 Geometry Library

In this section, the functions available in the geometry library of EMDLAB are explained.

2.1 emdlab_g2d_lib_tc1



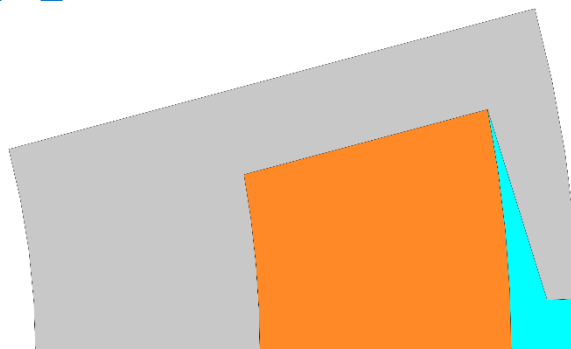
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. wt: tooth width: a double: constant tooth width
6. dss: depth of stator slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. name1: a name for iron region, for example: "stator": must be char or string.
11. name2: a name for coil arm region, for example: "sca": must be char or string.
12. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.2 emdlab_g2d_lib_tc2



Inputs:

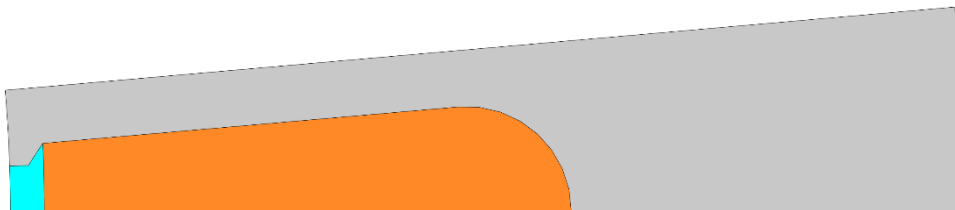
1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double

4. Ns: number of slots: an integer
5. wt: tooth width: a double: constant tooth width
6. dss: depth of stator slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. name1: a name for iron region, for example: "stator": must be char or string.
11. name2: a name for coil arm region, for example: "sca": must be char or string.
12. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.3 emdlab_g2d_lib_tc3



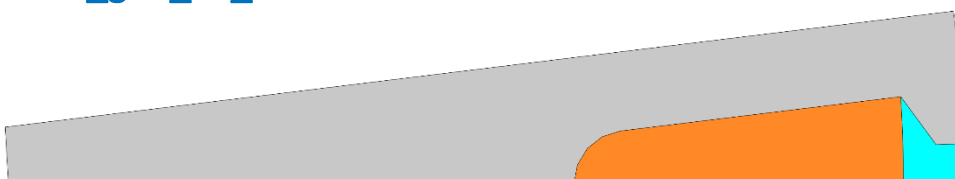
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. wt: tooth width: a double
6. dss: depth of stator slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. name1: a name for iron region, for example: "stator": must be char or string.
11. name2: a name for coil arm region, for example: "sca": must be char or string.
12. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.4 emdlab_g2d_lib_tc4



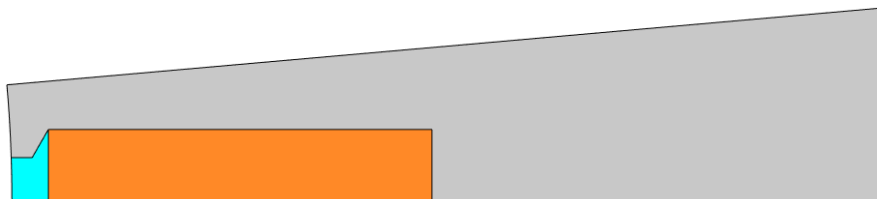
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. wt: tooth width: a double
6. dss: depth of stator slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. name1: a name for iron region, for example: "stator": must be char or string.
11. name2: a name for coil arm region, for example: "sca": must be char or string.
12. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.5 emdlab_g2d_lib_tc5



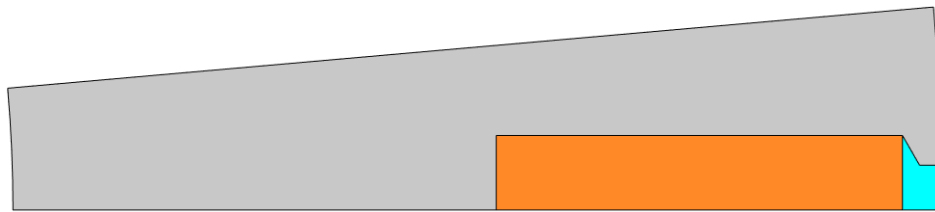
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. wslot: slot width: a double: constant slot width
6. dslot: depth of slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. name1: a name for iron region, for example: "stator": must be char or string.
11. name2: a name for coil arm region, for example: "sca": must be char or string.
12. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.6 emdlab_g2d_lib_tc6



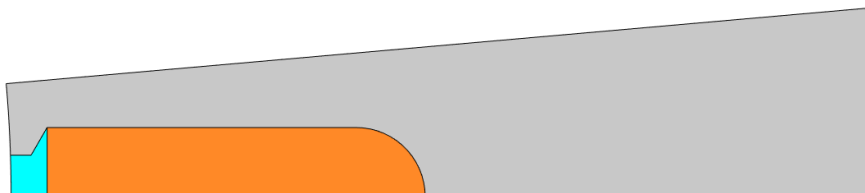
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. wslot: slot width: a double: constant slot width
6. dslot: depth of slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. name1: a name for iron region, for example: "rotor": must be char or string.
11. name2: a name for coil arm region, for example: "rca": must be char or string.
12. name3: a name for slot air region, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.7 emdlab_g2d_lib_tc7



Inputs:

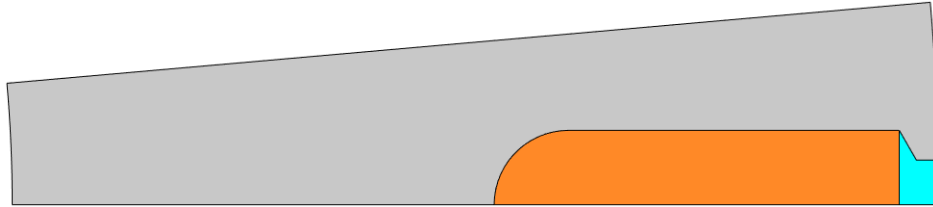
1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. wslot: slot width: a double: constant slot width with round bottom
6. dslot: depth of slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. name1: a name for iron region, for example: "stator": must be char or string.
11. name2: a name for coil arm region, for example: "sca": must be char or string.

12. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.8 emdlab_g2d_lib_tc8



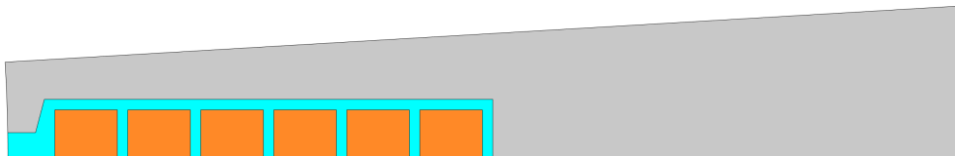
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. wslot: slot width: a double: constant slot width
6. dslot: depth of slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. name1: a name for iron region, for example: "rotor": must be char or string.
11. name2: a name for coil arm region, for example: "rca": must be char or string.
12. name3: a name for slot air region, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.9 emdlab_g2d_lib_tc9



Inputs:

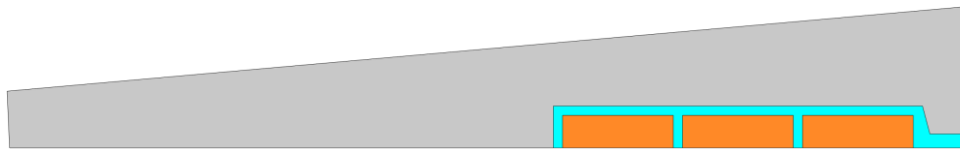
1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. Nc: number of conductors in each slot: an integer
6. wc: width of a conductor: a double

7. dc: depth of a conductor: a double
8. bs0: slot opening: a double
9. hs0: slot opening height: a double
10. tta: tooth tip angle: in degrees
11. d0x: distance between two conductors: a double
12. d0y: distance between one conductor and iron: a double
13. name1: a name for iron region, for example: "stator": must be char or string.
14. name2: a name for coil arm region, for example: "sca": must be char or string.
15. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.10emdlab_g2d_lib_tc10



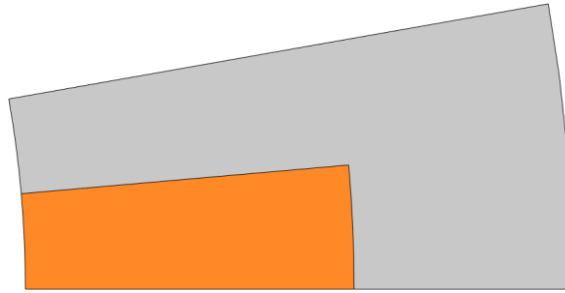
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. Nc: number of conductors in each slot: an integer
6. wc: width of a conductor: a double
7. dc: depth of a conductor: a double
8. bs0: slot opening: a double
9. hs0: slot opening height: a double
10. tta: tooth tip angle: in degrees
11. d0x: distance between two conductors: a double
12. d0y: distance between one conductor and iron: a double
13. name1: a name for iron region, for example: "stator": must be char or string.
14. name2: a name for coil arm region, for example: "sca": must be char or string.
15. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.11 emdlab_g2d_lib_tc11



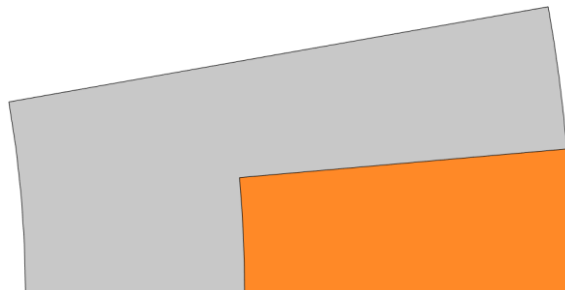
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. dslot: depth of a conductor: a double
6. ratio: ratio of slot arc per slot pitch angle
7. name1: a name for iron region, for example: "stator": must be char or string.
8. name2: a name for coil arm region, for example: "sca": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.12 emdlab_g2d_lib_tc12



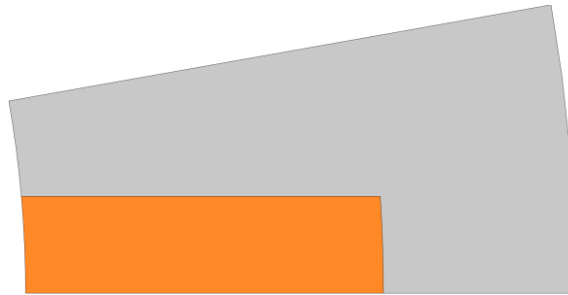
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. dslot: depth of a conductor: a double
6. ratio: ratio of slot arc per slot pitch angle
7. name1: a name for iron region, for example: "rotor": must be char or string.
8. name2: a name for coil arm region, for example: "rca": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.13 emdlab_g2d_lib_tc13



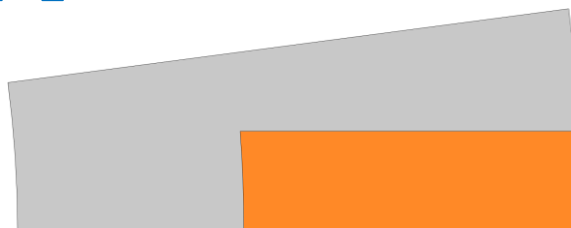
Inputs (constant slot width):

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. dslot: depth of a conductor: a double
6. wslot: slot width: a double
7. name1: a name for iron region, for example: "stator": must be char or string.
8. name2: a name for coil arm region, for example: "sca": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.14 emdlab_g2d_lib_tc14



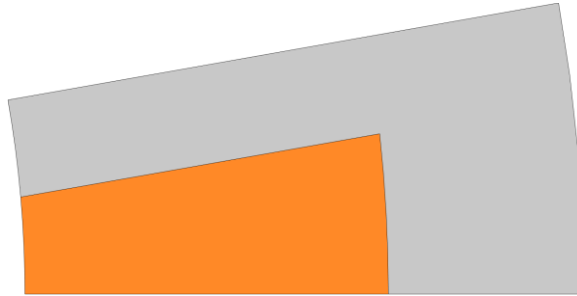
Inputs (constant slot width):

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. dslot: depth of a conductor: a double
6. wslot: slot width: a double
7. name1: a name for iron region, for example: "rotor": must be char or string.
8. name2: a name for coil arm region, for example: "rca": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.15emdlab_g2d_lib_tc15



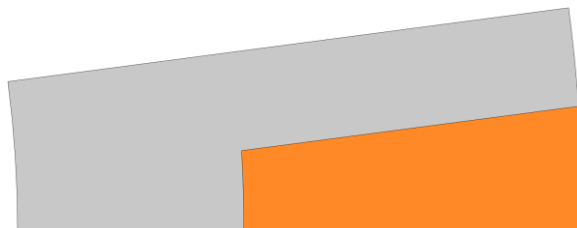
Inputs (constant tooth width):

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. dslot: depth of a conductor: a double
6. wtooth: tooth width: a double
7. name1: a name for iron region, for example: "stator": must be char or string.
8. name2: a name for coil arm region, for example: "sca": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.16emdlab_g2d_lib_tc16



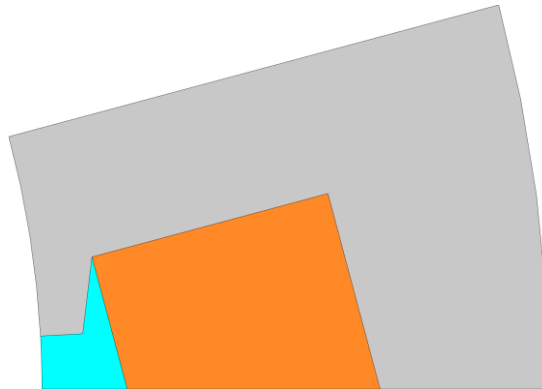
Inputs (constant tooth width):

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Nslots: number of slots: an integer
5. dslot: depth of a conductor: a double
6. wtooth: tooth width: a double
7. name1: a name for iron region, for example: "rotor": must be char or string.
8. name2: a name for coil arm region, for example: "rca": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.17emdlab_g2d_lib_tc17



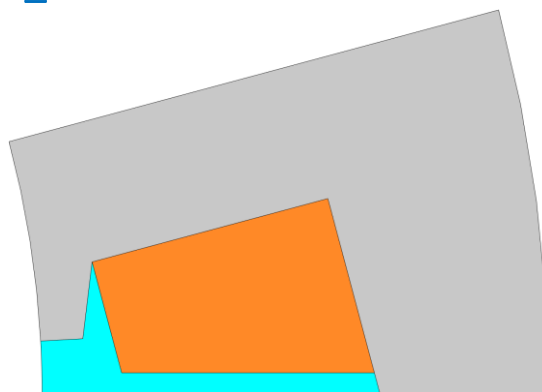
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. wt: tooth width: a double: constant tooth width
6. dss: depth of stator slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. name1: a name for iron region, for example: "stator": must be char or string.
11. name2: a name for coil arm region, for example: "sca": must be char or string.
12. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.18emdlab_g2d_lib_tc19



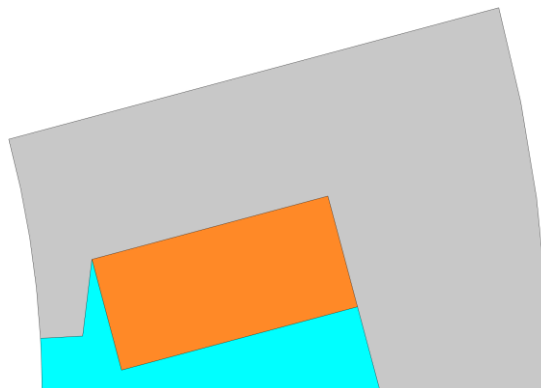
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. wt: tooth width: a double: constant tooth width
6. dss: depth of stator slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. bw0: distance between coil arms: a double: cannot be zero
11. name1: a name for iron region, for example: "stator": must be char or string.
12. name2: a name for coil arm region, for example: "sca": must be char or string.
13. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.19emdlab_g2d_lib_tc21



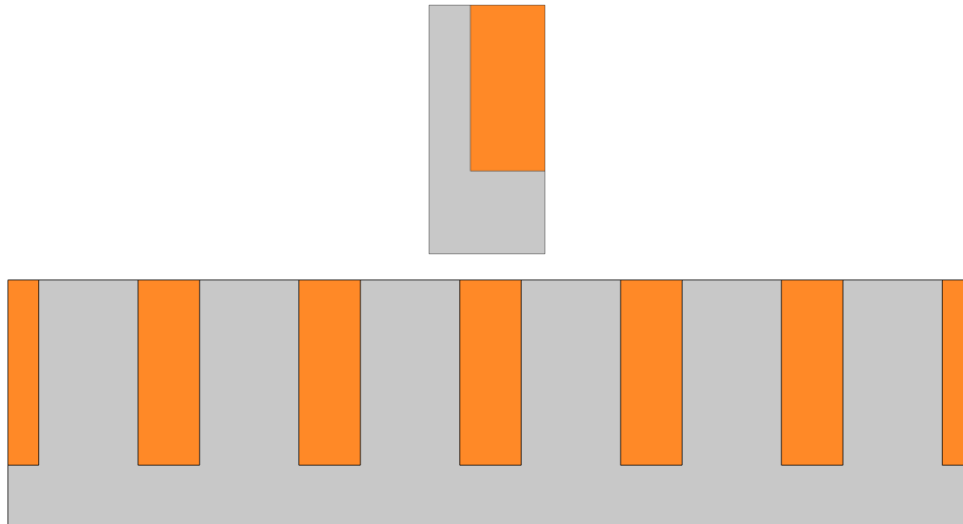
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. wt: tooth width: a double: constant tooth width
6. dss: depth of stator slot: a double
7. bs0: slot opening: a double
8. hs0: slot opening height: a double
9. tta: tooth tip angle: in degrees
10. bw0: distance between coil arms: a double: cannot be zero
11. name1: a name for iron region, for example: "stator": must be char or string.
12. name2: a name for coil arm region, for example: "sca": must be char or string.
13. name3: a name for slot air region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.20emdlab_g2d_lib_tc101



Inputs:

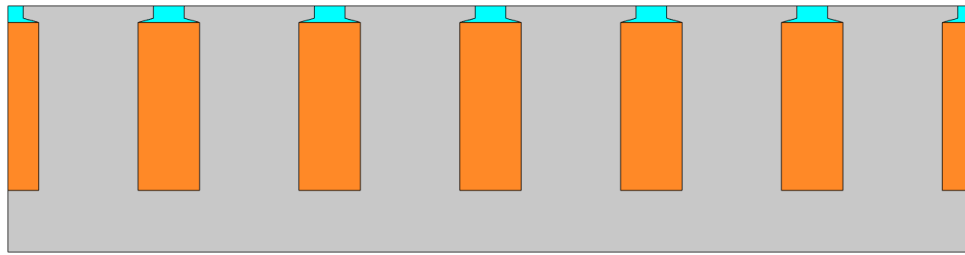
1. g: geometry database handle: emdlab_g2d_db
2. wst: tooth width: a double
3. wss: slot width: a double
4. dsy: yoke thickness: an integer
5. dss: slot depth: a double: constant tooth width
6. xShift: shift geometry in x-direction: a double
7. yShift: shift geometry in y-direction: a double
8. mirrorFlag: mirror geometry with respect to x-axis: true or false
9. name1: a name for iron region, for example: "stator": must be char or string.
10. name2: a name for coil arm region, for example: "sca": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.21emdlab_g2d_lib_tc102





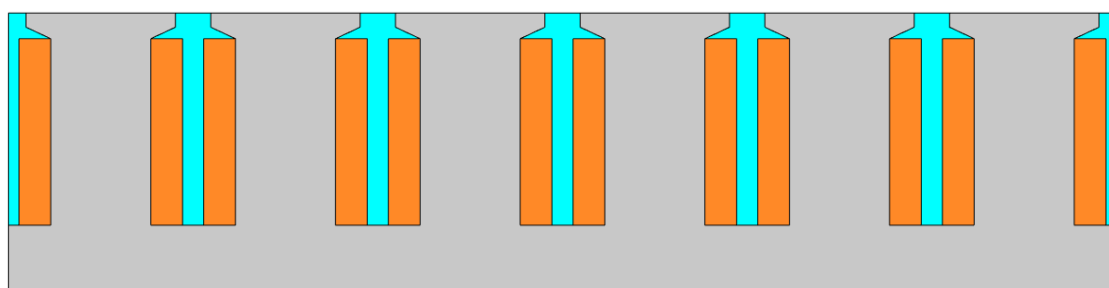
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. wst: tooth width: a double
3. wss: slot width: a double
4. dsy: yoke thickness: an integer
5. dss: slot depth: a double: constant tooth width
6. bs0: slot opening width: a double
7. hs0: slot opening height: a double
8. tta: tooth tip angle: in degrees
9. xShift: shift geometry in x-direction: a double
10. yShift: shift geometry in y-direction: a double
11. mirrorFlag: mirror geometry with respect to x-axis: true or false
12. name1: a name for iron region, for example: "stator": must be char or string.
13. name2: a name for coil arm region, for example: "sca": must be char or string.
14. name3: a name for stator air pocket region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.22emdlab_g2d_lib_tc103



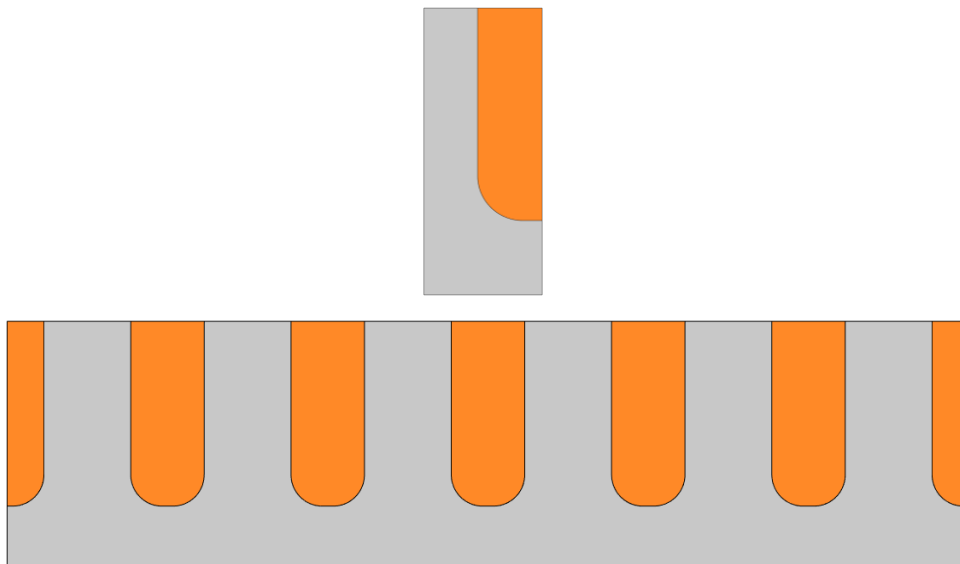
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. wst: tooth width: a double
3. wss: slot width: a double
4. dsy: yoke thickness: a double
5. dss: slot depth: a double: constant tooth width
6. bs0: slot opening width: a double
7. hs0: slot opening height: a double
8. tta: tooth tip angle: in degrees
9. bw0: coil to coil distance: a double
10. xShift: shift geometry in x-direction: a double
11. yShift: shift geometry in y-direction: a double
12. mirrorFlag: mirror geometry with respect to x-axis: true or false
13. name1: a name for iron region, for example: "stator": must be char or string.
14. name2: a name for coil arm region, for example: "sca": must be char or string.
15. name3: a name for stator air pocket region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.23emdlab_g2d_lib_tc105



Inputs:

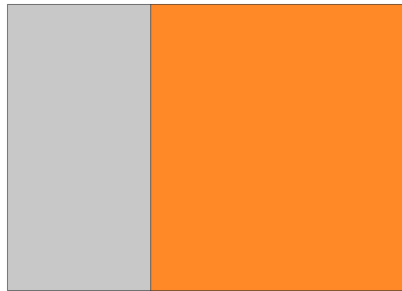
1. g: geometry database handle: emdlab_g2d_db
2. wst: tooth width: a double
3. wss: slot width: a double
4. dsy: yoke thickness: an integer
5. dss: slot depth: a double: constant tooth width

6. frr: fillet radius ratio: a double
7. xShift: shift geometry in x-direction: a double
8. yShift: shift geometry in y-direction: a double
9. mirrorFlag: mirror geometry with respect to x-axis: true or false
10. name1: a name for iron region, for example: "stator": must be char or string.
11. name2: a name for coil arm region, for example: "sca": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.24emdlab_g2d_lib_tc109



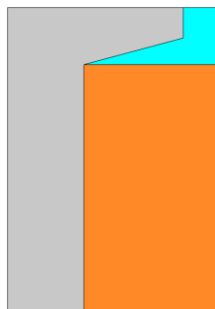
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. wst: tooth width: a double
3. wss: slot width: a double
4. dss: slot depth: a double
5. xShift: shift geometry in x-direction: a double
6. yShift: shift geometry in y-direction: a double
7. mirrorFlag: mirror geometry with respect to x-axis: true or false
8. name1: a name for iron region, for example: "stator": must be char or string.
9. name2: a name for coil arm region, for example: "sca": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

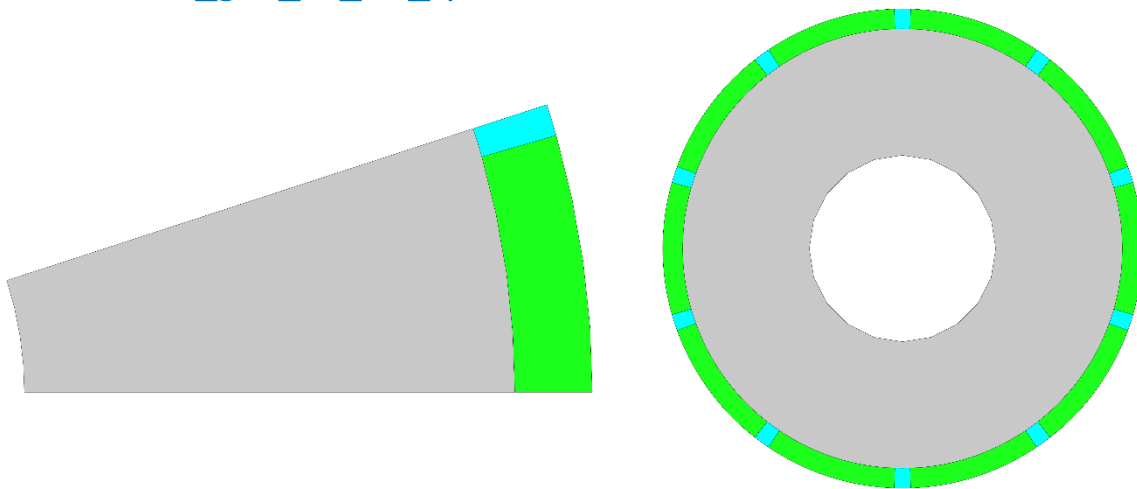
2.25emdlab_g2d_lib_tc110



Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. wst: tooth width: a double
3. wss: slot width: a double
4. dss: slot depth: a double: constant tooth width
5. bs0: slot opening width: a double
6. hs0: slot opening height: a double
7. tta: tooth tip angle: in degrees
8. xShift: shift geometry in x-direction: a double
9. yShift: shift geometry in y-direction: a double
10. mirrorFlag: mirror geometry with respect to x-axis: true or false
11. name1: a name for iron region, for example: "stator": must be char or string.
12. name2: a name for coil arm region, for example: "sca": must be char or string.
13. name3: a name for stator air pocket region, for example: "sap": must be char or string.

2.26emdlab_g2d_lib_rm_spm1



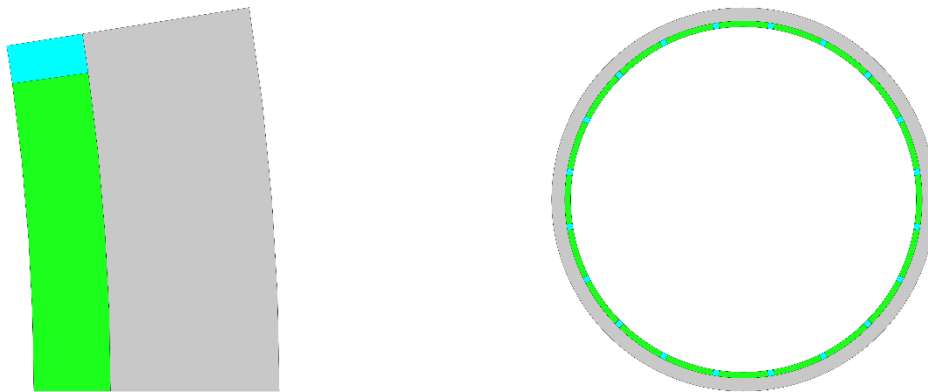
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm: magnet depth: a double
6. embrace: magnet embrace ratio: a double: must be lower than 1
7. name1: a name for iron region, for example: "rotor": must be char or string.
8. name2: a name for coil arm region, for example: "magnet": must be char or string.
9. name3: a name for slot air region, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.27 emdlab_g2d_lib_rm_spm2



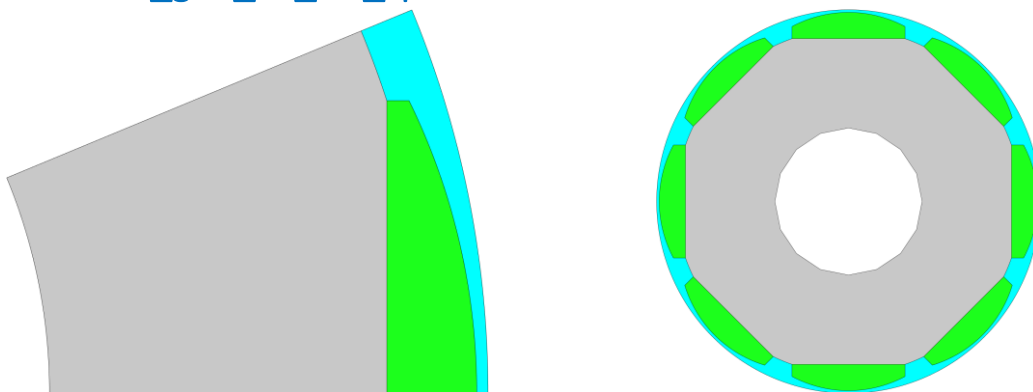
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm: magnet depth: a double
6. embrace: magnet embrace ratio: a double: must be lower than 1
7. name1: a name for iron region, for example: "rotor": must be char or string.
8. name2: a name for coil arm region, for example: "magnet": must be char or string.
9. name3: a name for slot air region, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.28 emdlab_g2d_lib_rm_spm3



Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. Dsh: shaft diameter: a double
3. ISD: inner stator diameter: a double
4. gap: air gap thickness: a double
5. gg: air gap ratio: a double greater than 1
6. Nagl: number of air gap mesh layers

7. p: number of poles: an integer
8. dm: magnet depth: a double
9. embrace: magnet embrace ratio: a double: must be lower than 1
10. name1: a name for iron region, for example: "rotor": must be char or string.
11. name2: a name for coil arm region, for example: "magnet": must be char or string.
12. name3: a name for slot air region, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.29emdlab_g2d_lib_rm_spm101



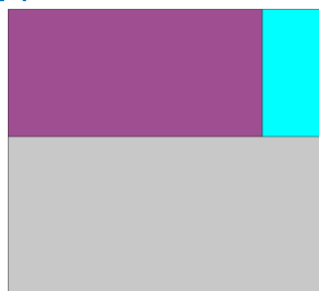
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. wpole: pole width: a double
3. dry: rotor yoke thickness: a double
4. dm: magnet depth: a double
5. xShift: shift geometry in x-direction: a double
6. yShift: shift geometry in y-direction: a double
7. mirrorFlag: mirror geometry with respect to x-axis: true or false
8. name1: a name for iron region, for example: "rotor": must be char or string.
9. name2: a name for coil arm region, for example: "magnet": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.30emdlab_g2d_lib_rm_spm102



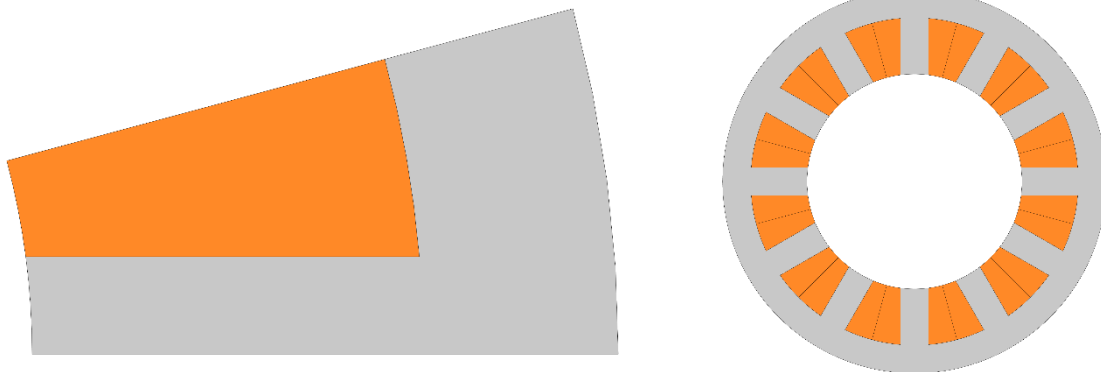
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. wpole: pole width: a double
3. dry: rotor yoke thickness: a double
4. dm: magnet depth: a double
5. embrace: magnet embrace ratio: must be lower than one
6. xShift: shift geometry in x-direction: a double
7. yShift: shift geometry in y-direction: a double
8. mirrorFlag: mirror geometry with respect to x-axis: true or false
9. name1: a name for iron region, for example: "rotor": must be char or string.
10. name2: a name for coil arm region, for example: "magnet": must be char or string.
11. name3: a name for air pocket region, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.31emdlab_g2d_lib_tc_srm1



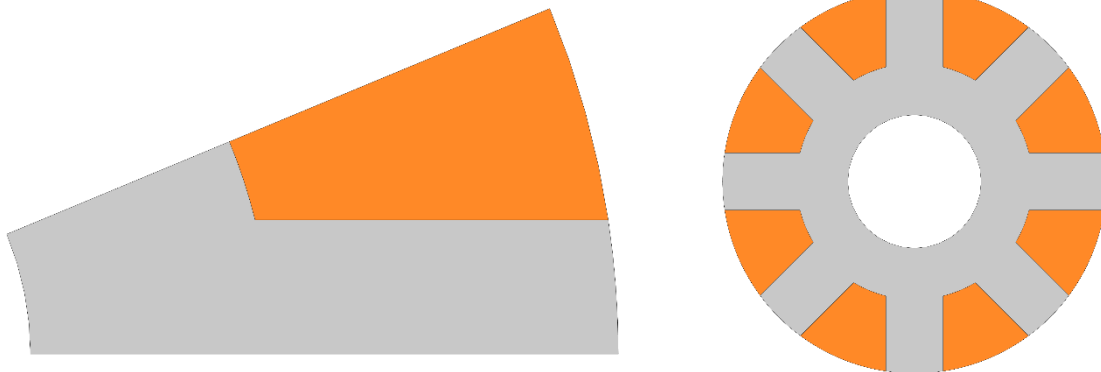
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. g_beta: pole arc angle per pole pith angle: a number lower than 1
6. g_ws: yoke width per tooth width: a number lower than 1
7. name1: a name for iron region, for example: "stator": must be char or string.
8. name2: a name for coil arm region, for example: "coil": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.32emdlab_g2d_lib_tc_srm2



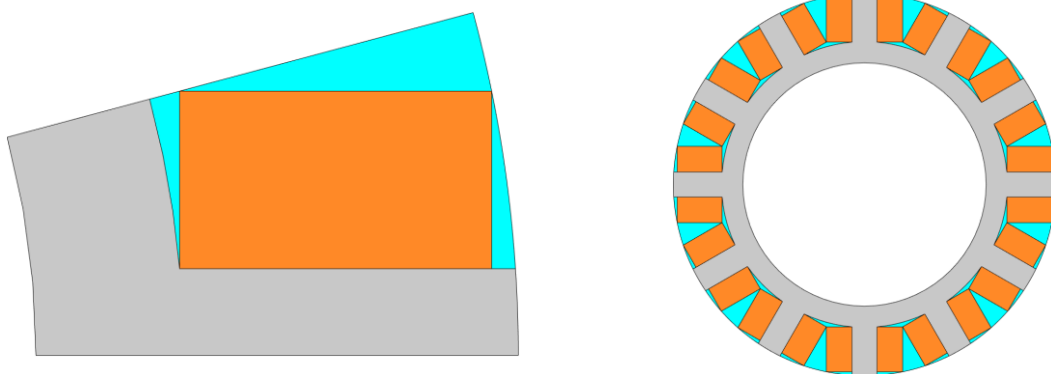
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. g_beta: pole arc angle per pole pith angle: a number lower than 1
6. g_ws: yoke width per tooth width: a number lower than 1
7. name1: a name for iron region, for example: "rotor": must be char or string.
8. name2: a name for coil arm region, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.33emdlab_g2d_lib_tc_srm4



Inputs:

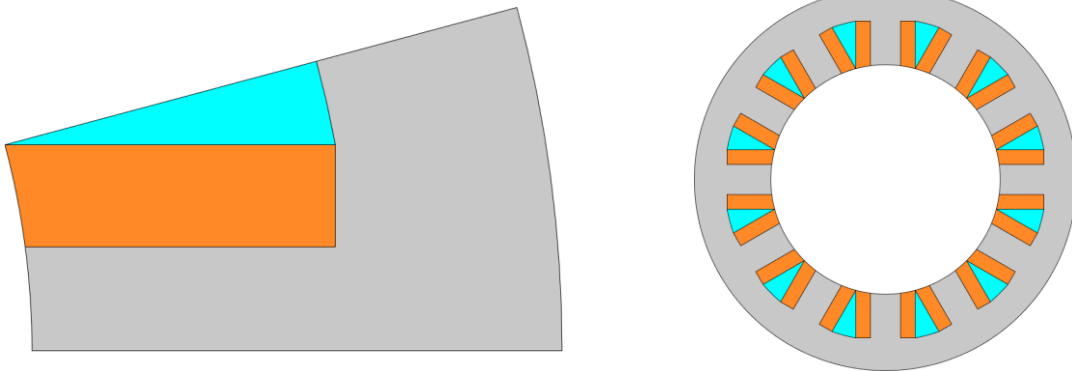
1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. g_beta: pole arc angle per pole pith angle: a number lower than 1
6. g_ws: yoke width per tooth width: a number lower than 1
7. name1: a name for iron region, for example: "rotor": must be char or string.

8. name2: a name for coil arm region, for example: "rca": must be char or string.
9. name3: a name for air pocket region, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.34emdlab_g2d_lib_tc_srm7



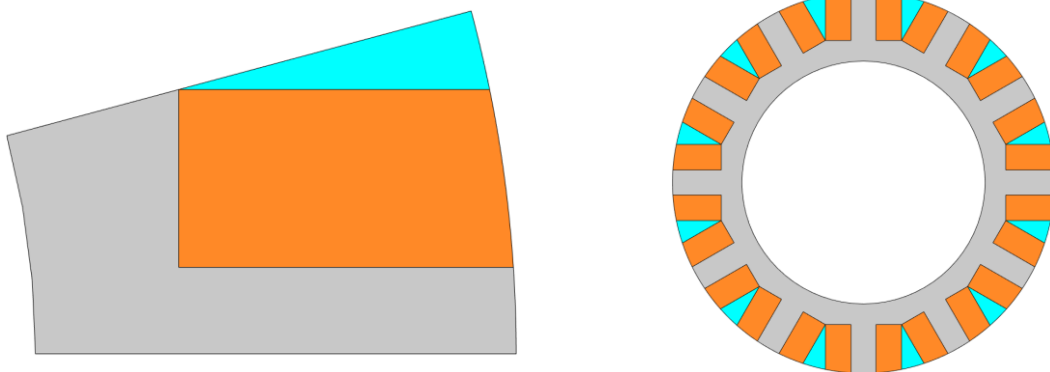
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. g_beta: pole arc angle per pole pith angle: a number lower than 1
6. g_ws: yoke width per tooth width: a number lower than 1
7. name1: a name for iron region, for example: "stator": must be char or string.
8. name2: a name for coil arm region, for example: "sca": must be char or string.
9. name3: a name for air pocket region, for example: "sap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.35emdlab_g2d_lib_tc_srm8



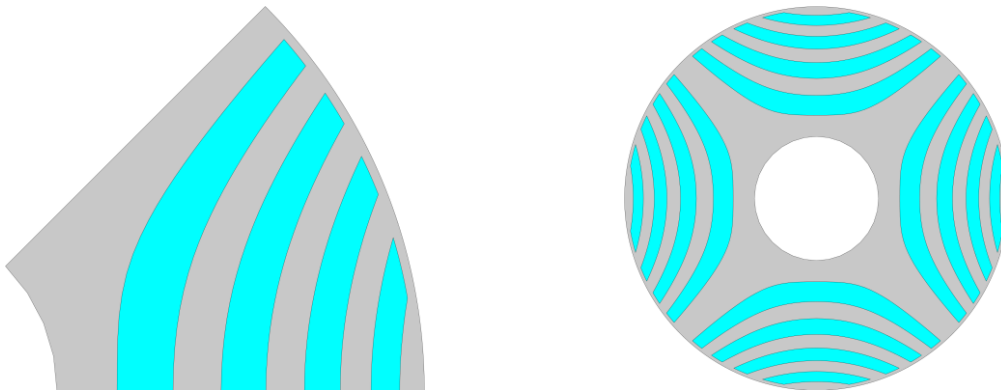
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. Ns: number of slots: an integer
5. g_beta: pole arc angle per pole pith angle: a number lower than 1
6. g_ws: yoke width per tooth width: a number lower than 1
7. name1: a name for iron region, for example: "rotor": must be char or string.
8. name2: a name for coil arm region, for example: "rca": must be char or string.
9. name3: a name for air pocket region, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.36emdlab_g2d_lib_rm_synrm1



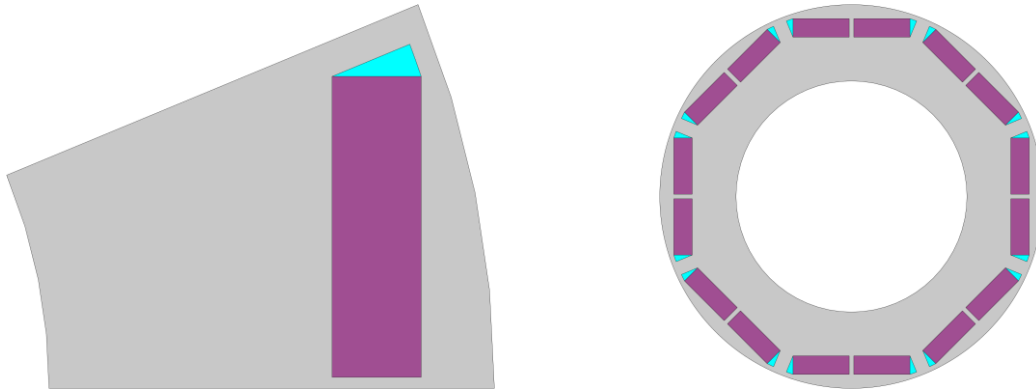
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. K_air: air insulation ratio: a number lower than 1
6. g_wy: an array of numbers lower than 1
7. g_db: an array of numbers lower than 1
8. name1: a name for iron region, for example: "rotor": must be char or string.
9. name2: a name for coil arm region, for example: "fb": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.37emdlab_g2d_lib_rm_ipm1



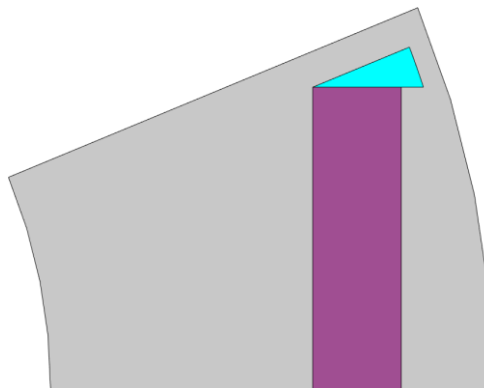
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm: depth of the magnet
6. wtrib: width of tangential rib: a double
7. wrrib: width of radial rib: a double
8. bm0: distance between magnets of adjacent poles
9. name1: a name for iron region, for example: "rotor": must be char or string.
10. name2: a name for magnet region, for example: "magnet": must be char or string.
11. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.38emdlab_g2d_lib_rm_ipm2



Inputs:

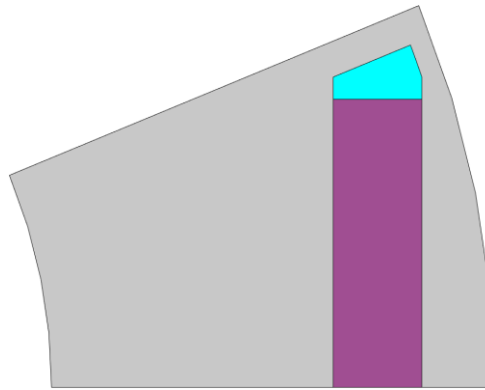
1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer

5. dm: depth of the magnet
6. wtrib: width of tangential rib: a double
7. wrrib: width of radial rib: a double
8. bm0: distance between magnets of adjacent poles
9. t0: distance between magnet and tangential rib: a double
10. name1: a name for iron region, for example: "rotor": must be char or string.
11. name2: a name for magnet region, for example: "magnet": must be char or string.
12. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.39emdlab_g2d_lib_rm_ipm3



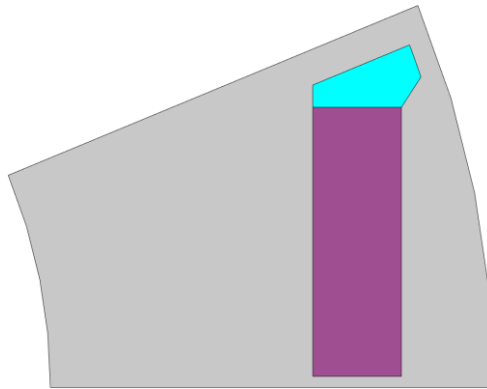
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm: depth of the magnet
6. wtrib: width of tangential rib: a double
7. wrrib: width of radial rib: a double
8. bm0: distance between magnets of adjacent poles
9. t0: distance between magnet and tangential rib: a double
10. name1: a name for iron region, for example: "rotor": must be char or string.
11. name2: a name for magnet region, for example: "magnet": must be char or string.
12. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.40emdlab_g2d_lib_rm_ipm4



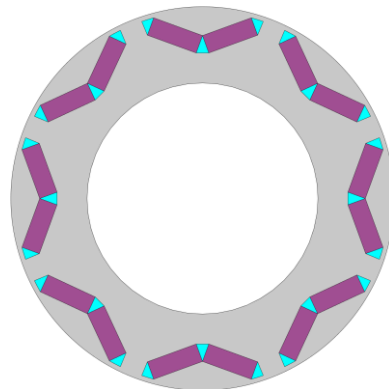
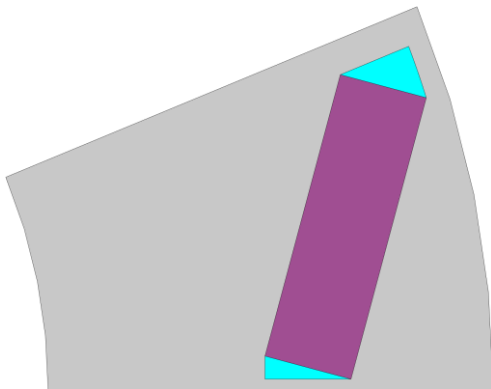
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm: depth of the magnet
6. wtrib: width of tangential rib: a double
7. wrrib: width of radial rib: a double
8. bm0: distance between magnets of adjacent poles
9. t0: horizontal distance between magnet and tangential rib: a double
10. t1: vertical distance between magnet and tangential rib: a double
11. name1: a name for iron region, for example: "rotor": must be char or string.
12. name2: a name for magnet region, for example: "magnet": must be char or string.
13. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.41emdlab_g2d_lib_rm_ipm5



Inputs:

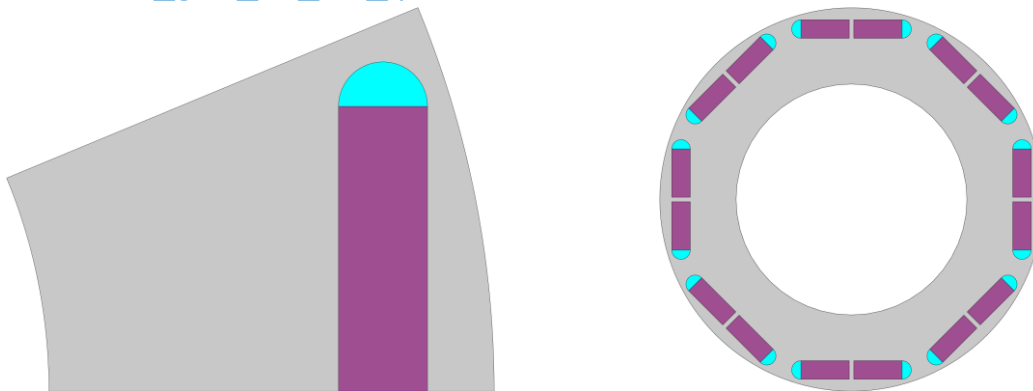
1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double

3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm: depth of the magnet
6. alpha_v: v-angle in degrees: a double
7. wtrib: width of tangential rib: a double
8. wrrib: width of radial rib: a double
9. bm0: distance between magnets of adjacent poles
10. name1: a name for iron region, for example: "rotor": must be char or string.
11. name2: a name for magnet region, for example: "magnet": must be char or string.
12. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.42emdlab_g2d_lib_rm_ipm9



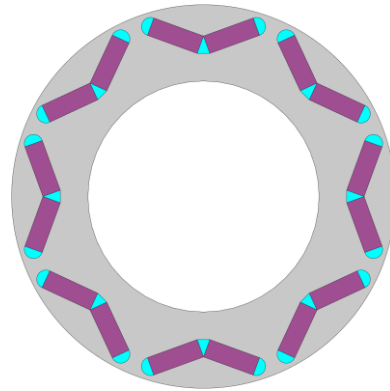
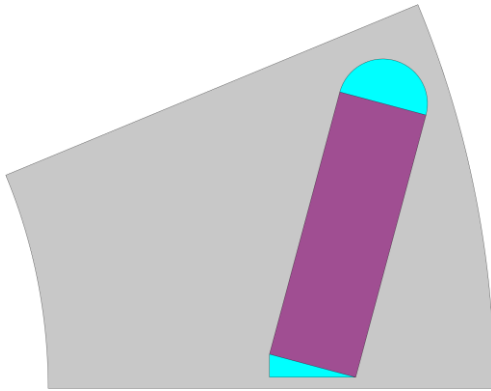
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm: depth of the magnet
6. wtrib: width of tangential rib: a double
7. wrrib: width of radial rib: a double
8. bm0: distance between magnets of adjacent poles
9. name1: a name for iron region, for example: "rotor": must be char or string.
10. name2: a name for magnet region, for example: "magnet": must be char or string.
11. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.43emdlab_g2d_lib_rm_ipm10



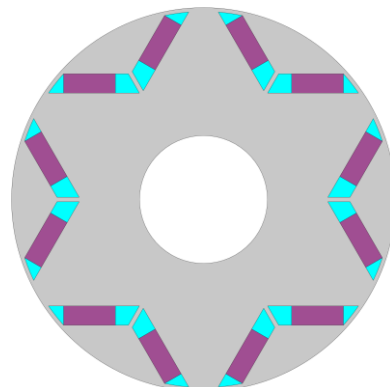
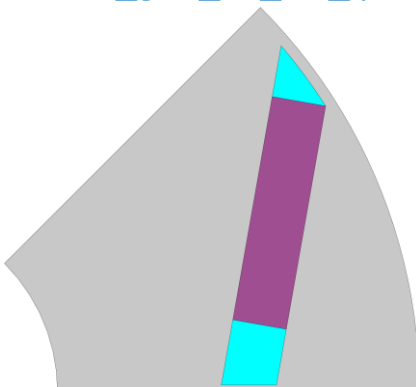
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm: depth of the magnet
6. alpha_v: v-angle in degrees: a double
7. wtrib: width of tangential rib: a double
8. wrrib: width of radial rib: a double
9. bm0: distance between magnets of adjacent poles
10. name1: a name for iron region, for example: "rotor": must be char or string.
11. name2: a name for magnet region, for example: "magnet": must be char or string.
12. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.44emdlab_g2d_lib_rm_ipm11



Inputs:

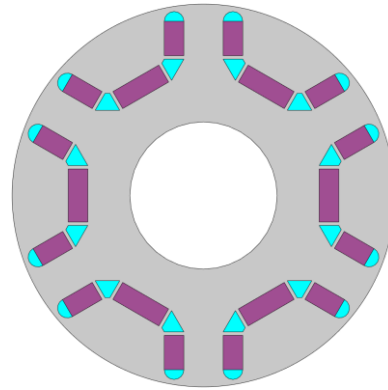
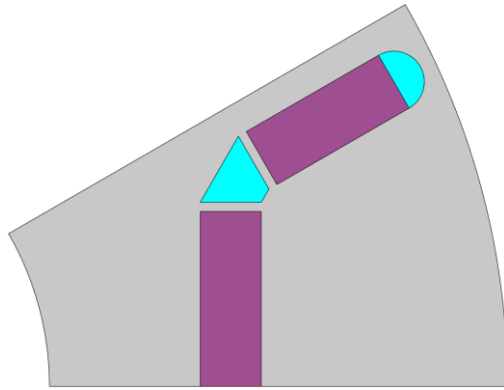
1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double

4. p: number of poles: an integer
5. alpha: ratio of v embrace to pole pitch angle: a double lower than 1
6. beta: the angle of magnet with respect to y-axis: in degrees
7. dm: depth of the magnet
8. gamma: fill factor of the magnet: a double lower than 1
9. wtrib: width of tangential rib: a double
10. wrrib: width of radial rib: a double
11. name1: a name for iron region, for example: "rotor": must be char or string.
12. name2: a name for magnet region, for example: "magnet": must be char or string.
13. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.45emdlab_g2d_lib_rm_ipm15



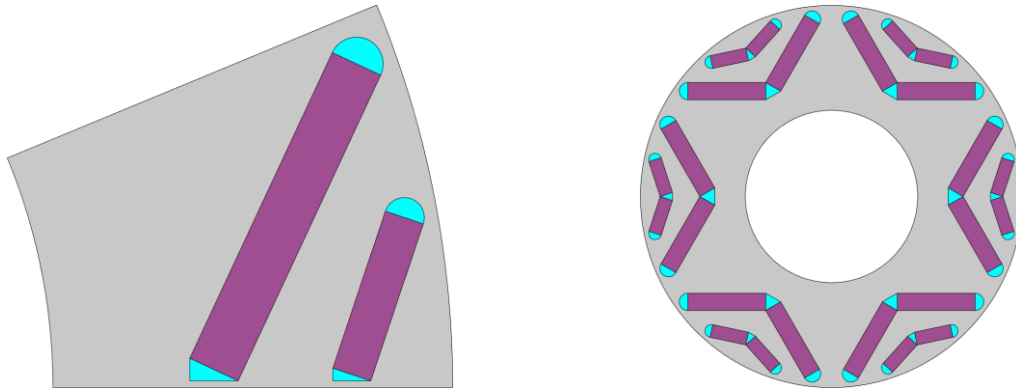
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm: depth of the magnet: a double
6. wsm: width of the side magnet: a double
7. w1: a double
8. w2: a double
9. w3: a double
10. w4: a double
11. name1: a name for iron region, for example: "rotor": must be char or string.
12. name2: a name for magnet regions, for example: "magnet": must be char or string.
13. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.46emdlab_g2d_lib_rm_ipm16



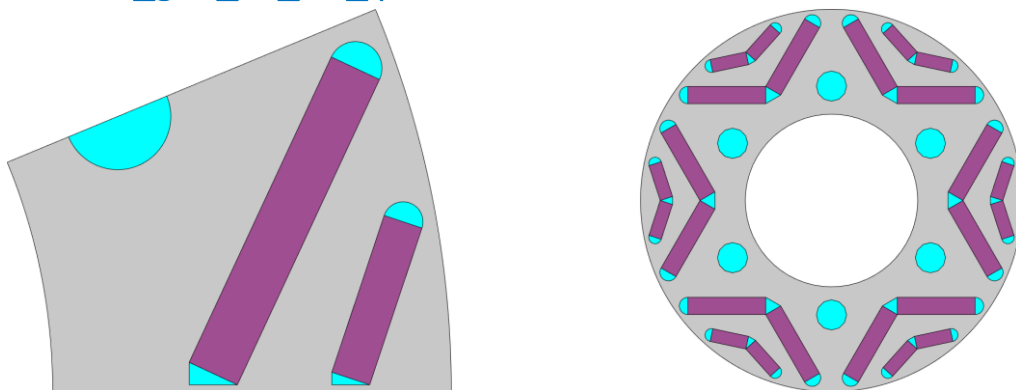
Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm1: depth of the inner magnet: a double
6. dm2: depth of the outer magnet: a double
7. alpha_v: inner v-angle in degrees
8. g_v: v-angle ratio
9. wtrib: width of tangential rib: a double
10. wrrib: width of radial rib: a double
11. d0: a double
12. d1: a double
13. name1: a name for iron region, for example: "rotor": must be char or string.
14. name2: a name for magnet regions, for example: "magnet": must be char or string.
15. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

2.47emdlab_g2d_lib_rm_ipm17



Inputs:

1. g: geometry database handle: emdlab_g2d_db
2. ID: inner diameter: a double
3. OD: outer diameter: a double
4. p: number of poles: an integer
5. dm1: depth of the inner magnet: a double
6. dm2: depth of the outer magnet: a double
7. alpha_v: inner v-angle in degrees
8. g_v: v-angle ratio
9. wtrib: width of tangential rib: a double
10. wrrib: width of radial rib: a double
11. d0: a double
12. d1: a double
13. d2: a double
14. hD: hole diameter: a double
15. name1: a name for iron region, for example: "rotor": must be char or string.
16. name2: a name for magnet regions, for example: "magnet": must be char or string.
17. name3: a name for air pocket regions, for example: "rap": must be char or string.

Outputs:

After running this function all related geometrical entities will be added to the geometry database.

3 EMDLAB Solvers

3.1 Magnetic Static

3.1.1 *emdlab_solvers_ms2d_tl3_ihn1*

This is a two-dimensional magnetic static solver in the xy-plane with following properties:

- first order triangular mesh
- triangular lagrangian elements: 3 points per element
- isotropic
- homogenous
- nonlinear solver: Newton-Raphson

3.1.2 *emdlab_solvers_ms2d_tl6_ihn1*

This is a two-dimensional magnetic static solver in the xy-plane with following properties:

- second order triangular mesh
- triangular lagrangian elements: 6 points per element
- isotropic
- homogenous
- nonlinear solver: Newton-Raphson

3.2 Magnetic Transinet

3.2.1 *emdlab_solvers_mt2d_tl3_ihnl_wtm*

This is a two-dimensional magnetic transient solver in the xy-plane with following properties:

- first order triangular mesh
- triangular lagrangian elements: 3 points per element
- isotropic
- homogenous
- nonlinear solver: Newton-Raphson
- without motion modeling

3.2.2 *emdlab_solvers_mt2d_tl3_ihnl_wm*

This is a two-dimensional magnetic transient solver in the xy-plane with following properties:

- first order triangular mesh
- triangular lagrangian elements: 3 points per element
- isotropic
- homogenous
- nonlinear solver: Newton-Raphson
- with motion modeling: re-mesh with fixed number of mesh nodes

3.3 Thermal Static

3.3.1 *emdlab_solvers_ts2d_tn*

This is a two-dimensional thermal static solver in the xy-plane based on thermal network theory.

3.3.2 *emdlab_solvers_ts3d_tn*

This is a three-dimensional thermal static solver based on thermal network theory.

3.4 Thermal Transient

3.4.1 *emdlab_solvers_tt2d_tn*

This is a two-dimensional thermal transient solver in the xy-plane based on thermal network theory.

3.4.2 *emdlab_solvers_tt3d_tn*

This is a three-dimensional thermal transient solver based on thermal network theory.

3.5 Optimization

3.5.1 *emdlab_solvers_opt_nsga2*

An optimization solver based on NSGA-II.